HDOC Day-51

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Write a C program that reads the size and elements of a variable-length integer array, arr[]. Pass the array to a function named next_greater() using its address. Use pointers to pass the array between the calling function and next_greater(). The function next_greater() must:

- Find the next greater element for each element in arr[].
- Search only the elements to the right of the current element and select the nearest element that is greater than the current element.
- Display -1 if no greater element exists to the right.
- Display the results in the order of their appearance in the array, separated by commas.
- Use nested loops.

1. Problem Understanding

The program reads the size and elements of a variable-length integer array. The array is passed to next_greater() using its address. For every element, the function searches only the elements to its right and selects the first (nearest) element that is greater. If no greater element exists, -1 is displayed. Nested loops are used as required.

2. Standard Test Cases

Test Case	Size (n)	Array Input	Expected Output
1	4	4 5 2 25	5, 25, 25, -1
2	4	13 7 6 12	-1, 12, 12, -1
3	4	1 3 2 4	3, 4, 4, -1
4	5	5 4 3 2 1	-1, -1, -1, -1, -1
5	5	2 2 3 1 4	3, 3, 4, 4, -1

3. Algorithm

Main Algorithm

1. Start.
2. Read the size n.
3. Declare a variable-length integer array arr[n].
4. Read n elements into the array.
5. Call next_greater(arr, n) by passing the array address.
6. Stop.

Function: next_greater()

1. Receive the array using an integer pointer and receive its size.
2. For each element at index i, set next = -1.
3. Use an inner loop from index i + 1 to the last element.
4. If the current right-side element is greater than arr[i], store it in next and break the inner loop.
5. Display next. Use commas between results.
6. Repeat for every element in the array.

4. C Program

```
#include <stdio.h>
```

```
void next_greater(int *arr, int n)
{
   int i, j, next;
   for(i = 0; i < n; i++)
   {
      next = -1;
      for(j = i + 1; j < n; j++)
      {
         if(*(arr + j) > *(arr + i))
         {
            next = *(arr + j);
            break;
         }
      }
      printf("%d", next);
      if(i < n - 1)
         printf(", ");
   }
}
int main()
{
   int n, i;
   printf("Enter the size of array: ");
   scanf("%d", &n);
   int arr[n];
   printf("Enter %d elements: ", n);
   for(i = 0; i < n; i++)
      scanf("%d", &arr[i]);
   printf("Next greater elements: ");
   next_greater(arr, n);
   printf("\n");
   return 0;
}
```

5. Compilation and Execution Evidence



6. Verification of All Test Cases

Test Case	Size (n)	Array Input	Actual Result	Status
1	4	4 5 2 25	5, 25, 25, -1	Pass
2	4	13 7 6 12	-1, 12, 12, -1	Pass
3	4	1 3 2 4	3, 4, 4, -1	Pass
4	5	5 4 3 2 1	-1, -1, -1, -1, -1	Pass
5	5	2 2 3 1 4	3, 3, 4, 4, -1	Pass